Safe AI Agents in Banking

Letting customers use AI helpers without handing over their passwords

Challenge: Banking & Finance | AIML-500 Workshop One Group Assignment

Group 2: Chandana Katta and four AIML-500 classmates

The problem: AI agents are logging in as customers

October 22, 2026: Morgan Stanley, JPMorgan, Bank of America and Citi found outside AI agents inside customer accounts, using the customers' own usernames and passwords.

- Some agents acted on their own: called bank call centers and did tasks customers never asked for.
- Most were not hackers. Customers gave their passwords to AI assistants.
- Banks can't tell a real customer from an agent, or whether the customer approved it.

Who is affected: customers, bank fraud teams, call centers, AI companies.

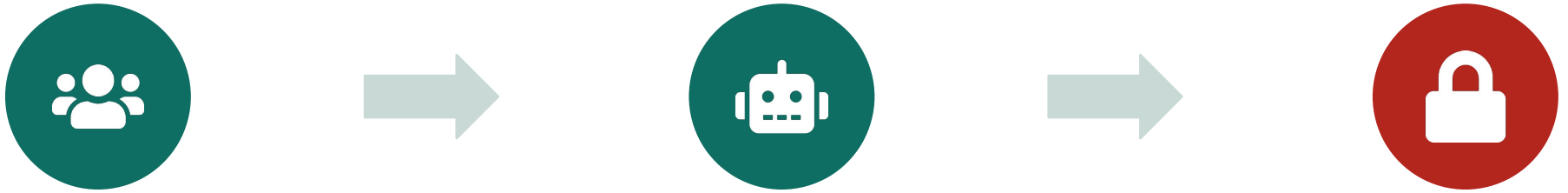
$15.9B lost to fraud by US consumers in 2025, up from $12B+ in 2024 (FTC)

78% of Americans already use AI tools in daily life (TD, 2026)

18% would trust AI to make financial recommendations on its own (TD, 2026)

Source: Axios via Yahoo Finance/IBT (Oct 22, 2026); Decrypt (Oct 22, 2026); FTC (Mar 2026); TD AI Insights Report (Mar 31, 2026)

Why it matters: one password unlocks everything



Customer

shares bank password

AI agent

logs in as the customer

Full access

view AND move money

No proof of approval
Bank can't confirm the customer allowed this agent or what it may do.

Hijack risk
Hidden text on a web page can trick an AI agent into following an attacker's commands (prompt injection, Brave 2025).

Who pays?
If an agent makes an unwanted transfer, liability is unclear. JPMorgan warns customers may bear it.

Blocking all agents hurts customers
People lose a useful tool, and real customers may be wrongly blocked.

Source: Yahoo Finance/IBT (Oct 22, 2026); Brave, "Indirect Prompt Injection in Perplexity Comet" (Aug 2025)

Proposed solution: AgentGuard, an AI trust layer

Don't let the AI impersonate the customer. Give the agent its own identity, limits, and approval path.



Verified agent identity

Agent connects through a bank-issued token, never the customer password.



Permission gate

Read-only by default; task-specific permissions and spending limits.



Intent check

Compares the customer's instruction with the action the agent is trying to do.



Risk engine

Classification + anomaly detection + prediction score each request.



Human control

Allow, ask, hold, or escalate. Sensitive money movement stays human-approved.

Relevant AI / ML concepts

Classification **Anomaly detection** **Prediction / risk** **NLP intent check**

Rules + permissions

Primary capability: classification + prediction. NLP supports intent matching; computer vision is not central.

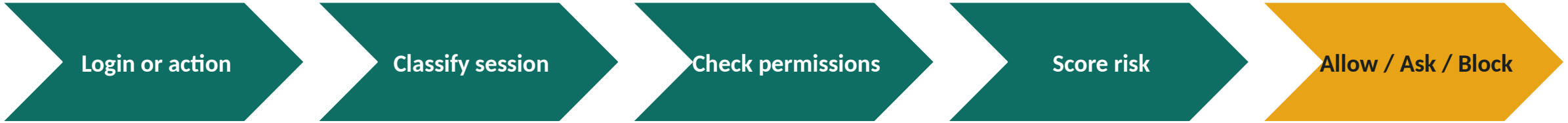
Benefits

- Safer AI delegation
- Clear audit trail
- Customer control
- Less password sharing

Limitations

False alarms, agent behavior drift, no shared integration standard yet, and added approval friction.

How it works: the AI and ML behind it



Classification
Human vs. approved agent vs. unknown agent, using typing speed, mouse and tap patterns, device and session signals.

Anomaly detection
Spots behavior that doesn't fit the customer or the agent's permissions, like a new payee plus a large transfer at 3 a.m.

Prediction (risk score)
Gives each action a 0–100 risk score. Low: allow. Medium: ask the customer. High: hold for a fraud analyst.

Language processing
Screens agent requests sent to call centers and chat for scam or hijack patterns.

Main AI capability: classification and prediction, trained on past sessions labeled human, agent, or fraud.

UX flow: safe delegation without losing control

A focused user flow for an AI agent preparing a bank payment.

1 Connect agent

Customer chooses Agent Guard inside the bank app.

2 Set permissions

Read balance? Prepare payments? New payees? Spending limit?

3 Agent requests action

AI prepares a $142 electricity payment.

4 Intent receipt

Bank shows exactly what the AI wants to do and why.

5 Risk check

Low → allow; medium → ask the customer; high → fraud review.

6 Activity center

Customer sees every agent action and can pause access anytime.

What we learned using generative AI

Our prompt (Claude, Anthropic)

"Act as a bank risk advisor. Customers are giving their online banking passwords to AI agents, and banks can't tell the agent from the customer. Suggest an AI-based solution, the ML methods involved, and its benefits and limitations."

Summary of the response (our words)

- Replace passwords with revocable, limited tokens.
- Use ML classification and anomaly detection to spot agents.
- Require customer approval for money movement.
- Limits: false alarms, agents changing behavior, no industry standard yet.

Fact we verified

AI claim: "Banks are moving agents to read-only access." Confirmed: Morgan Stanley is blocking outside agents; JPMorgan, BofA and Citi are restricting them and leaning toward read-only (Axios, via Decrypt, Oct 22, 2026).

How we checked the AI

- Matched each claim to news or government sources.
- Used numbers only from FTC and TD, not from the AI.
- Rejected "block all agents": simple, but unfair to customers.

Source: Decrypt, "Major Banks Find Rogue AI Agents Inside Customer Accounts" (Oct 22, 2026).

Using it responsibly: risks and safeguards



Accuracy

Risk: a real customer is blocked from paying rent, or a harmful agent gets through.

Safeguard: ask the customer first; never auto-close an account.



Bias & fairness

Risk: older adults, people using accessibility tools, and immigrants with unusual patterns get flagged more.

Safeguard: test error rates by group; fast appeal path.



Human oversight

Stays human: blocking money, closing accounts, fraud decisions, appeals.

AI only recommends; analysts decide.



Privacy & accountability

Protect: passwords, balances, behavior data. Never share passwords with agents.

Bank owns the detector; AI firms sign liability terms.

Biblical principles that shaped our design



Stewardship

"Whoever can be trusted with very little can also be trusted with much."
— Luke 16:10

Design choice: agents start read-only with small spending limits and earn more access only with the customer's approval.



Protecting the vulnerable

"Speak up for those who cannot speak for themselves… defend the rights of the poor and needy."
— Proverbs 31:8–9

Design choice: older and disabled customers get bias testing, a human review before any block, and a simple appeal.



Human dignity

"So God created mankind in his own image."
— Genesis 1:27

Design choice: a customer is a person, not a risk score. A human calls before any account is closed.

Source: Scripture quotations from the Holy Bible, New International Version (NIV)

Design thinking and a 90-day pilot

1

Empathize

Customers want AI help but fear losing money. Fraud teams can't see agents.

2

Define

Let approved agents help customers without passwords or unapproved money movement.

3

Ideate

AgentGuard: agent identity, permission gate, intent check, risk engine, human control.

4

Prototype

90-day pilot: 5,000 opted-in customers, 2 partner AI apps, read-only + bill pay up to $500.

5

Test

Measure success and harm (below). Adjust weekly.

Measure of success (target)

80%

fewer unapproved agent logins vs. the 90 days before the pilot

Unintended harm check (target)

≤ 1%

of real customers wrongly blocked, tracked by age group so no group is hit harder

Leading the change and our recommendation

Train staff

Fraud, call center and branch teams learn to spot agents, use risk scores, and handle appeals.

Communicate openly

Tell customers what agents can and can't do, how to connect safely, and never to share passwords.

Key leadership skill

Ethical judgment: knowing when AI should decide and when a person must.

Recommendation: PROCEED with a limited pilot

Customers are already using AI agents, so blocking them doesn't make the risk go away. A 90-day pilot with read-only defaults, spending limits, and human approval lets the bank learn safely. Expand only if both targets are met; revise or stop if real customers are harmed.



Thank you

Questions?

AgentGuard: safe AI agents in banking

Team: Chandana Katta and four AIML-500 classmates